

Exact Conditioning Order for Fabius Observation Masks

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Abstract

For independent common-tilt uniform coordinates, replacing an observed coordinate by one with a smaller cap decreases the information distinguishing the reference product law from any fixed bounded log-concave reweighting of the total sum. We prove convex ordering of the projected likelihood ratios, and hence binary Blackwell comparison and every f -divergence inequality. Common observations, overlapping masks, and countably many coordinates are permitted. The key is an elementary rectangle inequality for a function that is positive on one interval and nonpositive outside it. Applied to the original uniform product conditioned below a threshold, the result proves the earlier-versus-later mask comparison for every geometric ratio $0 < q < 1$ and every finite parameter value. The realizing binary kernel may depend on the chosen weight; no common kernel for all weights is asserted. All inequalities are nonstrict.

1 Model and comparison theorem

Let I be finite or countably infinite. Under Q , the coordinates are independent with densities

$$q_{\beta, a_i}(x) = \frac{e^{-\beta x}}{Z_\beta(a_i)} \mathbf{1}_{(0, a_i)}(x), \quad Z_\beta(a) = \int_0^a e^{-\beta x} dx, \quad (1)$$

where $\beta \in \mathbb{R}$, $a_i > 0$, and $A = \sum_i a_i < \infty$. Put $S = \sum_i X_i$. Fix a bounded nonnegative Borel log-concave function $w : \mathbb{R} \rightarrow [0, \infty)$ with

$$M = \mathbb{E}_Q w(S) > 0, \quad \frac{dP_w}{dQ} = \frac{w(S)}{M}. \quad (2)$$

Log-concavity allows zero values and interval supports. The expectation M is a global normalization; no positive conditional normalization will be required.

A mask $E \subseteq I$ observes $V_E = (X_i)_{i \in E}$. Its projected likelihood is

$$L_E = \mathbb{E}_Q \left[\frac{w(S)}{M} \middle| V_E \right]. \quad (3)$$

It is bounded, nonnegative, and has reference mean one. We use $\preceq_{\text{cx}}$ for convex order of real random variables.

Theorem 1.1 (Cap comparison for a fixed weight). *Suppose there is an injection $\pi : L \rightarrow E$ satisfying*

$$a_{\pi(j)} \geq a_j \quad (j \in L). \quad (4)$$

Masks may overlap, be empty or countably infinite, and have different cardinalities. Then

$$L_L \preceq_{\text{cx}} L_E \quad (5)$$

under their respective reference marginals. There is a Markov kernel taking both Q_E to Q_L and $(P_w)_E$ to $(P_w)_L$. In particular, for every convex divergence generator φ with $\varphi(1) = 0$,

$$D_\varphi((P_w)_E \| Q_E) \geq D_\varphi((P_w)_L \| Q_L), \quad (6)$$

with the usual extended-value convention. Therefore

$$\text{TV}((P_w)_E, Q_E) \geq \text{TV}((P_w)_L, Q_L), \quad \text{Overlap}_E \leq \text{Overlap}_L. \quad (7)$$

The binary kernel is allowed to depend on the fixed weight w .

The cap condition is sufficient for this binary comparison; no necessity or complete classification for a fixed weight is claimed. Strictness is also not asserted. For example, constant w gives $P_w = Q$ and equality for every mask.

2 The rectangle inequality

For a real number x , write $x_+ = \max(x, 0)$.

Lemma 2.1 (One positive interval). *Let $a \geq b > 0$, $L = a + b$, and let $g \in L^1([0, L])$. Suppose that, for some $0 \leq u \leq v \leq L$, $g \leq 0$ almost everywhere on $[0, u]$ and $[v, L]$, while $g \geq 0$ almost everywhere on $[u, v]$. Then*

$$\int_0^a \left(\int_0^b g(x+y) dy \right)_+ dx \geq \int_0^b \left(\int_0^a g(x+y) dx \right)_+ dy. \quad (8)$$

No zero-integral assumption is needed.

Proof. Let $G(x) = \int_0^x g(s) ds$. It is absolutely continuous and is nonincreasing, then nondecreasing, then nonincreasing. For $0 < h < L$, put

$$J(h) = \int_0^{L-h} [G(x+h) - G(x)]_+ dx.$$

The elementary layer-cake identity gives

$$J(h) = \int_{\mathbb{R}} \int_0^{L-h} (\mathbf{1}_{\{G(x+h) > t\}} - \mathbf{1}_{\{G(x) > t\}})_+ dx dt. \quad (9)$$

Tonelli applies; G is bounded on its compact domain.

At each level t , the superlevel set of G is, up to endpoints, an initial interval together with at most one later interval. Write it as

$$[0, p] \cup [q, r], \quad 0 \leq p \leq q \leq r \leq L,$$

allowing empty or degenerate intervals. This representation follows directly from the three monotone pieces of G . Plateaus at level t belong to the complement; any plateau between the two superlevel components is included in $[p, q]$. Thus no exceptional-level argument is needed.

A positive indicator increment starts in the gap $[p, q]$ and ends in the later interval $[q, r]$. Its length is

$$f(h) = |[p, q] \cap [q-h, r-h]|.$$

The restriction to $[0, L-h]$ is automatic for such endpoints. With $u_0 = q - p$ and $v_0 = r - q$,

$$f(h) = \max(0, \min(h, u_0, v_0, u_0 + v_0 - h)). \quad (10)$$

Here $u_0 + v_0 = r - p \leq a + b$. If $f(a) > 0$, then $f(a)$ is at most u_0 , v_0 , and $u_0 + v_0 - a$. The last quantity is at most both b and $u_0 + v_0 - b$, because $u_0 + v_0 \leq a + b$ and $a \geq b$. Therefore $f(a) \leq f(b)$. The same conclusion is immediate if $f(a) = 0$.

Integrating this levelwise comparison in (9) gives $J(b) \geq J(a)$, which is exactly (8). $\square$

3 From the rectangle to likelihood convex order

Proposition 3.1 (Two coordinates). *Let X, Y be independent with the densities (1) on caps $a \geq b$. Let $h \geq 0$ be bounded and have interval superlevel sets. Set*

$$H_X(x) = \mathbb{E}h(x + Y), \quad H_Y(y) = \mathbb{E}h(X + y).$$

Then $H_Y(Y) \preceq_{\text{cx}} H_X(X)$. This includes $h \equiv 0$.

Proof. The variables have the same mean $\mathbb{E}h(X + Y)$. For a stop-loss level $t \geq 0$, put

$$g_t(s) = e^{-\beta s}(h(s) - t), \quad 0 \leq s \leq a + b.$$

The positive set of g_t is an interval, so Lemma 2.1 applies. Writing $Z_a = Z_\beta(a)$ and $Z_b = Z_\beta(b)$, positive homogeneity of x_+ gives the exact identity

$$\mathbb{E}(H_X(X) - t)_+ = \frac{1}{Z_a Z_b} \int_0^a \left(\int_0^b e^{-\beta(x+y)}(h(x+y) - t) dy \right)_+ dx. \quad (11)$$

The target has the reversed rectangle integral. Thus its stop-loss expectation is no larger for every $t \geq 0$. At negative t , equality follows from nonnegativity and equal means. The stop-loss characterization of convex order proves the result. $\square$

3.1 Hidden and common coordinates

Compare observing (V_C, X) with observing (V_C, Y) , where C is any common coordinate mask disjoint from the two coordinates and all remaining coordinates are hidden. Let R be their hidden sum. Conditional on $V_C = v$, put $c = \sum_{i \in C} v_i$ and define

$$h_c(s) = \frac{\mathbb{E}_R w(c + s + R)}{M}. \quad (12)$$

This is bounded and log-concave in s . For finitely many hidden coordinates, it follows by applying Prékopa's marginalization theorem [1, Theorem 6] to

$$w \left(c + s + \sum_j r_j \right) \prod_j q_{\beta, a_j}(r_j).$$

Each factor is log-concave, including the interval supports. If there are no hidden coordinates, (12) is just a translate of w divided by M .

For countably many hidden coordinates, their finite partial sums converge almost surely to R . A nonempty hidden sum has a density, since it contains one independent absolutely continuous coordinate. A bounded nonzero log-concave function is continuous in the interior of its interval support and can have discontinuities only at its endpoints. For each fixed s , bounded convergence therefore gives the pointwise limit of the finite effective profiles. Pointwise limits preserve log-concavity. A profile may be identically zero for some c ; that conditional comparison is trivial and requires no division by its mean.

The projected likelihoods conditional on $V_C = v$ are precisely $H_X(X)$ and $H_Y(Y)$ from Proposition 3.1, with $h = h_c$. Apply (11) for each v and integrate against the shared reference law Q_C . This proves convex-order comparison of the full likelihood ratios for the cap-decreasing exchange, with all common observations included.

3.2 Finite and countable mask comparisons

For finite masks of equal size satisfying (4), the matching may be chosen to fix every common index. To see this, if a common index j is matched from $i \neq j$ and is itself used as the source for another target k , replace these two assignments by $j \mapsto j$ and $i \mapsto k$. The inequalities $a_i \geq a_j \geq a_k$ preserve admissibility. If j was unused as a source, match it to itself and free i . Repeating fixes all common indices.

The remaining source and target indices are disjoint. Replace one matched source by its target at a time, treating every other currently observed coordinate as common. The preceding exchange comparison applies at each step for the same fixed pair (P_w, Q) . Extra source observations can be discarded by conditional Jensen. This proves (5) for all finite target masks.

For a countably infinite target mask, take increasing finite prefixes L_k and their matched finite source sets $E_k \subseteq E$. Then, for every real t ,

$$\mathbb{E}_Q(L_{L_k} - t)_+ \leq \mathbb{E}_Q(L_{E_k} - t)_+ \leq \mathbb{E}_Q(L_E - t)_+.$$

The bounded likelihood martingale L_{L_k} converges in L^1 to L_L . Since stop-loss functions are Lipschitz, the inequality passes to the limit. This proves (5) for arbitrary masks in the theorem.

3.3 The binary kernel and divergences

For completeness, convex ordering yields the asserted kernel through the classical martingale-coupling theorem of Strassen [2]. Let ξ have the reference law of L_E and η that of L_L . There is a coupling with

$$\mathbb{E}[\xi \mid \eta] = \eta.$$

Given a source observation, compute its likelihood ξ , draw η from this coupling conditional on ξ , then draw a target observation from Q_L conditional on its likelihood being η . These conditional kernels exist on the finite or countable product Borel spaces used here. Under Q_E the output law is Q_L . Under $(P_w)_E$, the coupling is weighted by ξ ; conditioning on η changes that weight to η , which gives exactly $(P_w)_L$. Null-set versions are immaterial because $P_w \ll Q$.

Finally, $D_\varphi((P_w)_E \| Q_E) = \mathbb{E}_Q \varphi(L_E)$, so convex order gives (6), or equivalently one may use the resulting binary kernel and data processing. The same kernel compares divergences in the reverse orientation. Since $\text{TV}(P, Q) = \mathbb{E}_Q(L - 1)_+$ and overlap is $1 - \text{TV}$, equation (7) follows. This completes the proof of Theorem 1.1.

4 Original uniform conditioning and the Fabius masks

Let P_0 be the original product of uniforms on the caps, and let Q_β be its common exponential tilt:

$$\frac{dQ_\beta}{dP_0} = \frac{e^{-\beta S}}{\mathbb{E}_{P_0} e^{-\beta S}}.$$

For any threshold τ with $P_0(S \leq \tau) > 0$, the conditioned law $P_0(\cdot \mid S \leq \tau)$ is a reweighting of Q_β by

$$w(s) = e^{\beta s} \mathbf{1}_{[0, \tau]}(s).$$

The total is nonnegative, so the lower cutoff does not change the law. This weight is bounded and log-concave. Theorem 1.1 therefore applies for every real tilt, every admissible threshold, and every cap-dominating mask injection.

A direct check of the effective profile. In this application, on the relevant range $s \geq 0$, (12) is a positive constant times

$$e^{\beta s} F_R^0(\tau - c - s), \quad (13)$$

where F_R^0 is the hidden-sum CDF under the original uniform product. For k hidden coordinates, the sublevel sets

$$B_t = \{r \in \prod_{j=1}^k [0, a_j] : \sum_j r_j \leq t\}$$

satisfy $(1 - \lambda)B_s + \lambda B_t \subseteq B_{(1-\lambda)s + \lambda t}$. Brunn–Minkowski followed by the arithmetic-geometric mean inequality makes their volume, and hence F_R^0 , log-concave. For infinitely many hidden coordinates, the partial-sum CDFs decrease pointwise to F_R^0 , preserving log-concavity. For no hidden coordinates the CDF is a half-line indicator. Thus (13) supplies a direct verification for the actual conditioning rule.

To match the Fabius source convention exactly, let

$$S_q = \sum_{j \geq 1} q^j U_j, \quad U_j \stackrel{\text{iid}}{\sim} \text{Unif}[0, 1], \quad t_n = \rho q^{-(n+1)}, \quad T_n = t_n S_q,$$

and define

$$\frac{dQ_n}{dP_0} = \frac{e^{-T_n}}{\mathbb{E}_{P_0} e^{-T_n}}, \quad \mu_n = \mathbb{E}_{Q_n} T_n, \quad P_n = P_0(\cdot \mid T_n \leq \mu_n).$$

Under Q_n , $X_j = t_n q^j U_j$ has cap $a_j = \rho q^{j-n-1}$ and tilt $\beta = 1$. The two masks in the critical-complement comparison are

$$B_{n,m}^{\text{late}} = \{1, \dots, n - m\}, \quad B_{n,m}^{\text{early}} = \{m + 1, \dots, n\}. \quad (14)$$

The superscript names the *hidden* bulk block. Both masks also hide the entire infinite boundary tail. The late-hidden mask observes the larger, earlier caps.

Corollary 4.1 (Every geometric ratio and finite parameter). *For every $q \in (0, 1)$, $\rho > 0$, $n \geq 2$, and $1 \leq m < n$, the binary experiment (P_n, Q_n) observed on $B_{n,m}^{\text{late}}$ Blackwell-dominates that observed on $B_{n,m}^{\text{early}}$. In particular,*

$$\text{Overlap}_{n,m}^{\text{late}} \leq \text{Overlap}_{n,m}^{\text{early}}. \quad (15)$$

All f -divergences have the corresponding opposite information-loss order. Common observations outside the two masks may be included in both.

Proof. Pair the r th source index r with the target $m + r$. Its source cap is larger by the factor $q^{-m} > 1$. Apply Theorem 1.1 to the bounded log-concave weight $e^{T_n} \mathbf{1}_{\{T_n \leq \mu_n\}}$. Common observations are handled by adjoining the same indices to both masks and retaining an admissible cap matching. $\square$

5 Scope and verification

The kernel in Theorem 1.1 may depend on w . There is generally no single kernel valid for every log-concave weight: lower-threshold indicators are themselves log-concave, and one kernel for every threshold already determines the entire sum-reweighting experiment. The arithmetic obstructions proved in [4] therefore remain valid. The present theorem settles a different comparison, the binary pair for each fixed log-concave weight.

Arbitrary additive noise is not included without a further hypothesis. The proof requires interval superlevel sets for the effective conditional profile; a log-concave hidden-sum law is sufficient. The ordinary finite or countable truncated-exponential sums in the stated model

meet this condition. No strict inequality is claimed; equality and strictness for the particular Fabius thresholds remain separate questions.

The source’s arbitrary-mask question [5] concerns sharper cap-profile-dependent crossover behavior. The finite-parameter order (15) complements those asymptotic questions. It strengthens the comparison of the two specified masks beyond the eventual critical-window comparison in [3], while using their exact observed-versus-hidden convention.

The package contains exact rational checks of the rectangle inequality for piecewise-constant sign profiles, together with reproducible finite-product numerical checks. These support the formulas and orientation; the proofs above establish the continuous and countable statements. Prékopa marginalization, Brunn–Minkowski, convex order, and martingale coupling are classical inputs. The report makes no worldwide priority claim and is an ordinary mathematical proof, not a formalized or externally refereed result.

References

- [1] A. Prékopa, *On logarithmic concave measures and functions*, Acta Scientiarum Mathematicarum (Szeged) **34** (1973), 335–343, especially Theorems 6–8.
https://acta.bibl.u-szeged.hu/14411/1/math_034_335-343.pdf.
- [2] V. Strassen, *The existence of probability measures with given marginals*, Annals of Mathematical Statistics **36** (1965), 423–439. doi:10.1214/aoms/1177700153.
- [3] *Second Order Critical Complements in Fabius Conditioning*, research companion prepared for Vladimir Reshetnikov with OpenAI, 1 October 2026. Source `Second_Order_Fabius_Crossover.tex`.
- [4] *Arithmetic Classification of Geometric Observation Masks*, research companion prepared for Vladimir Reshetnikov with OpenAI, 1 October 2026. Source `arithmetic_geometric_mask_order.tex`.
- [5] V. Reshetnikov, ProveIt, *Critical Complements in Fabius Conditioning*, source at commit 7421a4ca60fd, inspected 1 October 2026. Pinned source on GitHub. Full identifiers and the exact question scope are recorded in the package.